Algebra 2

Group Activity — Answer Key

Unit 1, Lesson 1.2

Name: ____________________ Partner: ____________________ Period: ________

How Many Answers Does This Question Have? An equation is a question, and the honest answer is sometimes "every number" and sometimes "none." With your partner, decide *what kind of question* each item is before you touch it — then answer it, and check. Argue about the ones you disagree on before writing anything down.

Tier R — Remediate

Part 1 — Solve, and name one move. Solve each equation. In the last column, name *one* property of equality you used.

Equation	Solution	One property you used
(a) $3x + 7 = 22$	$x = 5$	**subtraction, then division property of equality**
(b) $2(x - 4) = 10$	$x = 9$	**distributive property, then addition and division**
(c) $5x - 3 = 2x + 12$	$x = 5$	**subtraction property of equality (subtract $2x$)**
(d) $\frac{x}{4} + 2 = 7$	$x = 20$	**multiplication property of equality (times 4)**

Pick any one and check it by substituting into the original. Which one, and does it check? **e.g. (c):** $5(5)-3 = 22$ **and** $2(5) + 12 = 22$ **— yes**

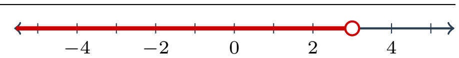
Part 2 — Read the last line only. Four students finished four different equations. You are given only their *final* line. Say how many solutions each equation has. Do not solve anything.

Final line	How many solutions?	How you knew
(a) $x = 3$	**one**	**the variable survived**
(b) $5 = 5$	**infinitely many**	**variable gone, statement true**
(c) $0 = 7$	**none**	**variable gone, statement false**
(d) $-2 = -2$	**infinitely many**	**variable gone, statement true**

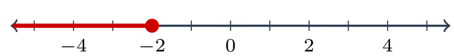
In (b), (c), and (d) the variable disappeared. Is that a mistake? **No** What decides which answer you give? **whether the leftover statement is true or false**

Part 3 — Solve and shade. Solve, then shade the solution set. Watch your circle: open ∘ for $<$ and $>$, closed • for $\leq$ and $\geq$.

(a) $2x + 1 < 7$ $x < 3$



(b) $-4x \geq 8$ $x \leq -2$



One of those two required you to reverse the symbol. Which, and why? **(b) — both sides were divided by -4, a negative**

Part 4 — Quadratics that factor. Every one of these factors cleanly. Set one side to **zero** first if it is not already.

(a) $x^2 - 9 = 0$, $x = \pm 3$ (b) $x^2 + 6x + 8 = 0$, $x = -2,\ -4$

(c) $x^2 - 7x + 12 = 0$, $x = 3,\ 4$ (d) $x^2 + 5x = 0$, $x = 0,\ -5$

Check yourself. Substitute *one* of your answers to (b) into $x^2 + 6x + 8$. Do you get zero? **At $x = -2$:** $4 - 12 + 8 = 0$ **— yes**

Every quadratic here had *two* answers. Item (a) is the one people get half right — why? $x^2 = 9$ **has two roots,** 3 **and** -3**; only the positive one gets written**

Tier A — Approaching Proficiency

Part 1 — Mixed, and nobody is telling you which kind it is. Decide first, then work.

1. $\dfrac{2x}{3} - 1 = \dfrac{x}{2} + 2$

$$\begin{aligned} \frac{2x}{3} - 1 &= \frac{x}{2} + 2 \\ 4x - 6 &= 3x + 12 \\ x - 6 &= 12 \\ x &= 18 \end{aligned}$$

   (Multiply every term by 6 first.)

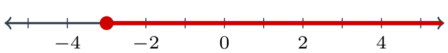
2. Solve for x: $\quad ax + 3x = b$.

$$\begin{aligned} ax + 3x &= b \\ x(a+3) &= b \\ x &= \frac{b}{a+3} \end{aligned}$$

   The variable appears *twice*. What did you have to do before you could divide? **factor** x **out — and note** $a \neq -3$

3. $8 - 2x \leq 14$

$$\begin{aligned} 8 - 2x &\leq 14 \\ -2x &\leq 6 \\ x &\geq -3 \end{aligned}$$



4. $x^2 = 4x + 21$

$$\begin{aligned} x^2 &= 4x + 21 \\ x^2 - 4x - 21 &= 0 \\ (x-7)(x+3) &= 0 \\ x &= 7 \quad \text{or} \quad x = -3 \end{aligned}$$

5. $x^2 + 6x + 4 = 0$ — nothing factors. Use the formula and **simplify the radical**.

$$\begin{aligned} x &= \frac{-6 \pm \sqrt{6^2 - 4(1)(4)}}{2(1)} \\ &= \frac{-6 \pm \sqrt{20}}{2} \\ &= \frac{-6 \pm 2\sqrt{5}}{2} \\ &= -3 \pm \sqrt{5} \end{aligned}$$

   Is your answer rational or irrational? **irrational**

Part 2 — Two situations, answered in sentences.

6. A parking garage charges \$4 to enter plus \$2.50 per hour. You have \$20.

   Inequality: $4 + 2.5h \le 20$ Solve: $h \le 6.4$

   Answer the question that was asked, in a sentence: **You can park for at most 6.4 hours — 6 hours and 24 minutes — before the charge passes \$20. (Also: $h \ge 0$, since negative hours mean nothing here.)**

7. A ball is thrown upward. Its height in feet after t seconds is $-16t^2 + 64t$. When is the ball 48 feet high?

   Equation: $-16t^2 + 64t = 48$ $t =$ **1 and 3 seconds**

   You got *two* answers, and both are real times. Explain what is physically happening at each one. **The ball passes 48 feet twice: at $t = 1$ s on the way up and again at $t = 3$ s on the way back down. Two solutions is the honest answer, not a redundancy to discard.**

Part 3 — Predict, then solve. For $2x^2 - 4x - 3 = 0$:

$b^2 - 4ac = 40$ Prediction: **two** real solutions Now solve: $x = \dfrac{2 \pm \sqrt{10}}{2}$

Did the count match your prediction? **yes** Why is predicting first worth anything at all? **Because it is a check you can run before you have anything to check — if the formula hands back one answer when the discriminant said two, you know a sign got lost. It also tells you immediately when to stop: a negative discriminant means there is nothing to find over the reals.**

Tier E — Extension

Part 1 — Put the parameter inside the equation. Until now the number of solutions was something you discovered *after* solving. Make it a condition instead.

Number of solutions	For which k does $kx + 6 = 3x + 6$?	For which k does $kx + 6 = 3x + 9$?
exactly one	**every $k \ne 3$ (then $x = 0$)**	**every $k \ne 3$**
no solution	**no such k**	**$k = 3$ (gives $0 = 3$)**
infinitely many	**$k = 3$**	**no such k**

One box in each column has *no* possible k. Say which, and why that case simply cannot happen for that equation. **Both equations reduce to $(k-3)x =$ constant — to 0 on the left, to 3 on the right. The first can never be a contradiction: the only way the x terms cancel is $k = 3$, and then the two sides are *identical* ($3x + 6$ and $3x + 6$), so the leftover statement $0 = 0$ is automatically true. The second can never be an identity: when $k = 3$ the x terms cancel but the constants still disagree, leaving $0 = 3$. Which case is unreachable is decided entirely by whether the constants match once the variable is gone.**

Part 2 — Spot the error. Each student made **exactly one** mistake. **Name** what went wrong — naming it is the part that counts — and give the correct answer.

	Student's work	What went wrong (name it)	Correct
(a)	$-2x > 10 \Rightarrow x > -5$	**divided by a negative without reversing the symbol**	$x < -5$
(b)	$(x-3)(x+2) = 6$, so $x - 3 = 6$ or $x + 2 = 6$	**zero product property used when the product is not zero**	$x = 4, -3$
(c)	$x^2 = 25 \Rightarrow x = 5$	**dropped the negative root — half the solution set**	$x = \pm 5$
(d)	$3(x-4) = 3x - 12$: "the x's cancel, so no solution"	**read an identity as a contradiction**	**all reals**

Two of these four are failures to *interpret* rather than failures to compute. Which two? **(b) and (d)**

Part 3 — The discriminant as a condition. Consider $x^2 - 6x + c = 0$, where c can be any real number.

(a) Write the discriminant in terms of c: $36 - 4c$

(b) Two real solutions when $c < 9$; one real solution when $c = 9$; no real solutions when $c > 9$

(c) You just answered a question about *equations* by solving an *inequality*. Say in one sentence why that was the right tool. **"Two real solutions" is the condition $36 - 4c > 0$ — a statement about a whole *range* of c values, and describing a range is exactly what an inequality is for.** . . . .

Part 4 — Why zero and nothing else. The zero product property says $mn = 0$ forces $m = 0$ or $n = 0$. There is no version of it for $mn = 8$. Explain why not — and use the word **factors**. **Eight can be split into two nonzero factors in infinitely many ways — $8 \cdot 1$, $4 \cdot 2$, $16 \cdot \frac{1}{2}$, $-2 \cdot -4$ — so knowing the product is 8 tells you nothing about either factor on its own. Zero is the only real number that cannot be built from two nonzero factors, so if a product is zero at least one factor is forced to be zero. That uniqueness is the whole property, and it is why Step 0 of factoring is always "make one side zero."** . . . .

Teacher Note

Tier R Part 2 is the highest-leverage block on the page. Students classify without solving, which strips the task down to the only thing that matters — reading the leftover statement. Anyone who called $0 = 7$ "zero solutions" but $5 = 5$ "one solution" has the idea half-formed; that pair is the diagnostic.

Tier A item 7 is worth taking to the whole room even if only some students reach it. Two solutions to a height equation are not an artifact — the ball genuinely is at 48 feet twice. Students who have been trained that "the second answer gets thrown out" need to hear that the throwing-out happens only when the context forbids it (negative time, negative lengths), not because two answers are suspicious.

Tier E Part 1 is the hardest thing on the page and the most worthwhile. It converts the classification from an observation into a condition on a coefficient — which is how the idea will be used from Unit 3 onward. Expect the "no such k" boxes to be left blank; the point is that $kx + 6 = 3x + 6$ *cannot* be a contradiction, because matching x terms here also means matching constants.

Tier E Part 3 is the first time both halves of the lesson run at once — a quadratic question answered with a linear inequality. If a group finishes early, ask what happens to $x^2 - 6x + c = 0$ as c passes through 9, and connect it to the parabola sliding up until it just touches the x-axis and then clears it. That picture is Unit 3's opening move.